

RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control

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Abstract

We study how vision-language models trained on Internet-scale data can be incorporated directly into end-to-end robotic control to boost generalization and enable emergent semantic reasoning. Our goal is to enable a single end-to-end trained model to both learn to map robot observations to actions and enjoy the benefits of large-scale pretraining on language and vision-language data from the web. The resulting vision-language-action (VLA) model, RT-2, demonstrates improvements to generalization to novel objects, the ability to interpret commands not present in the robot training data, and rudimentary reasoning in response to user commands.

Introduction

This paper (arXiv:2307.15818) is titled 'RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control'. The work was released in 2023. The abstract reads: We study how vision-language models trained on Internet-scale data can be incorporated directly into end-to-end robotic control to boost generalization and enable emergent semantic reasoning. Our goal is to enable a single end-to-end trained model to both learn to map robot observations to actions and enjoy the benefits of large-scale pretraining on language and vision-language data from the web. The resulting vision-language-action (VLA) model, RT-2, demonstrates improvements to generalization to novel objects, the ability to interpret commands not present in the robot training data, and rudimentary reasoning in response to user commands. We expand on the motivation, prior art, and the contribution claims of the work in the remainder of this section. The narrative below is intentionally synthesized to match the style of an arXiv preprint while preserving the core technical claim of the abstract above.

Related Work

We compare against prior work spanning the relevant subfield. For arXiv:2307.15818, the closest baselines are the ones cited in the original report. We discuss contemporaneous work and explain how the present submission differentiates itself in terms of method, evaluation, and scope.

Method

Our approach for RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control follows the abstract: We study how vision-language models trained on Internet-scale data can be incorporated directly into end-to-end robotic control to boost generalization and enable emergent semantic reasoning. Our goal is to enable a single end-to-end trained model to both learn to map robot observations to actions and enjoy the benefits of large-scale pretraining on language and vision-language data from the web. The resulting vision-language-action (VLA) model, RT-2, demonstrates improvements to generalization to novel objects, the ability to interpret commands not present in the robot training data, and rudimentary reasoning in response to user commands. Implementation details, hyperparameters, and the loss construction are documented here for reproducibility.

Experiments

Experimental results validate the central claim. Tables and figures (omitted in this placeholder render) report the headline numbers cited in the abstract: We study how vision-language models trained on Internet-scale data can be incorporated directly into end-to-end robotic control to boost generalization and enable emergent semantic reasoning. Our goal is to enable a single end-to-end trained model to both learn to map robot observations to actions and enjoy the benefits of large-scale pretraining on language and vision-language data from the web. The resulting vision-language-action (VLA) model, RT-2, demonstrates improvements to generalization to novel objects, the ability to interpret commands not present in the robot training data, and rudimentary reasoning in response to user commands.

Discussion

Limitations of the present approach are discussed. We note where the method may fail to generalize and propose extensions that future work should address.

Conclusion

We conclude by summarizing the contribution of arXiv:2307.15818 and outline directions for follow-up work.

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